EASY · ALGEBRA

# Linear equations in two variables

03

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

**Q1** GRID-IN ALGEBRA

Vivian bought party hats and cupcakes for $\$ 71$. Each package of party hats cost $\$ 3$, and each cupcake cost $\$ 1$. If Vivian bought $10$ packages of party hats, how many cupcakes did she buy?

STUDENT-PRODUCED RESPONSE

| | | | |
|---|---|---|---|
| . | / . | / . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 ALGEBRA

Last week, an interior designer earned a total of $\$ 1,258$ from consulting for $x$ hours and drawing up plans for $y$ hours. The equation $68x + 85y = 1,258$ represents this situation. Which of the following is the best interpretation of $68$ in this context?

A. The interior designer earned $\$ 68$ per hour consulting last week.

B. The interior designer worked $68$ hours drawing up plans last week.

C. The interior designer earned $\$ 68$ per hour drawing up plans last week.

D. The interior designer worked $68$ hours consulting last week.

Q3 ALGEBRA

A gardener buys two kinds of fertilizer. Fertilizer A contains 60% filler materials by weight and Fertilizer B contains 40% filler materials by weight. Together, the fertilizers bought by the gardener contain a total of 240 pounds of filler materials. Which equation models this relationship, where x is the number of pounds of Fertilizer A and y is the number of pounds of Fertilizer B?

A. $0.4x + 0.6y = 240$

B. $0.6x + 0.4y = 240$

C. $40x + 60y = 240$

D. $60x + 40y = 240$

Q4 ALGEBRA

A producer is creating a video with a length of 70 minutes. The video will consist of segments that are 1 minute long and segments that are 3 minutes long. Which equation represents this situation, where $x$ represents the number of 1-minute segments and $y$ represents the number of 3-minute segments?

A. $4xy = 70$

B. $4x + y = 70$

C. $3x + y = 70$

D. $x + 3y = 70$

Q5 ALGEBRA

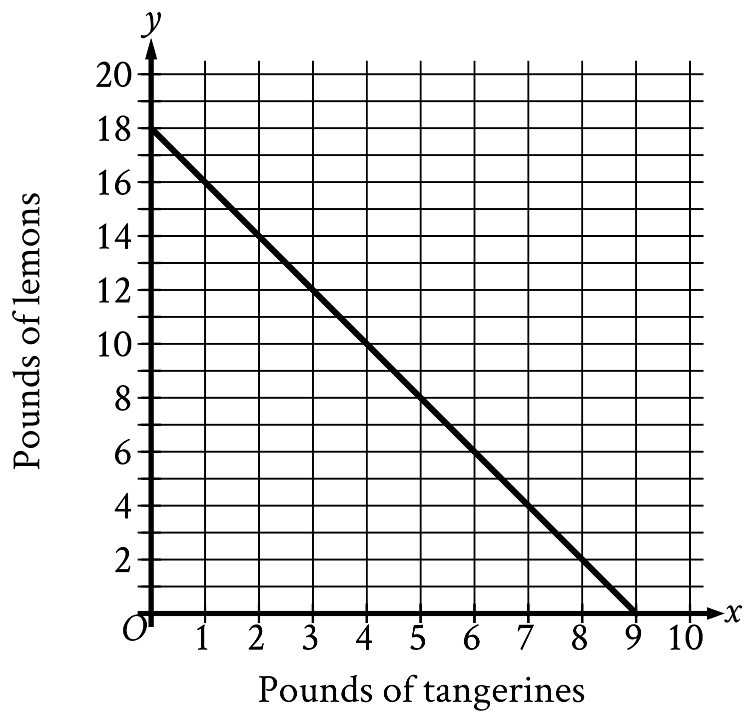
In a chess tournament, each participant earns 1 point for each game the participant plays that ends in a draw and 3 points for each game the participant wins. A certain participant in this tournament has earned 41 points. Which equation represents this situation, where $d$ represents the number of games this participant has played that ended in a draw and $w$ represents the number of games this participant has won?

A. $d + 3w = 41$

B. $3d + w = 41$

C. $d + \frac{w}{3} = 41$

D. $\frac{d}{3} + w = 41$




- The line slants sharply down from left to right.
- The line passes through the following points:
  - (0 comma 18)
  - (4 comma 10)
  - (7 comma 4)
  - (9 comma 0)

The graph shows the possible combinations of the number of pounds of tangerines and lemons that could be purchased for $\$18$ at a certain store. If Melvin purchased lemons and 4 pounds of tangerines for a total of $\$18$, how many pounds of lemons did he purchase?

**A.** 7

**B.** 10

C. 14

D. 16

Q7 ALGEBRA

$$y = x + 4$$

Which table gives three values of $x$ and their corresponding values of $y$ for the given equation?

A.

| $x$ | $y$ |
|---|---|
| 0 | 4 |
| 1 | 5 |
| 2 | 6 |

B.

| $x$ | $y$ |
|---|---|
| 0 | 6 |
| 1 | 5 |
| 2 | 4 |

C.

| $x$ | $y$ |
|---|---|
| 0 | 2 |
| 1 | 1 |
| 2 | 0 |

D.

| $x$ | $y$ |
|---|---|
| 0 | 0 |
| 1 | 1 |
| 2 | 2 |

Q8 ALGEBRA

$$x + y = 75$$

The equation above relates the number of minutes, $x$, Maria spends running each day and the number of minutes, $y$, she spends biking each day. In the equation, what does the number 75 represent?

**A.** The number of minutes spent running each day

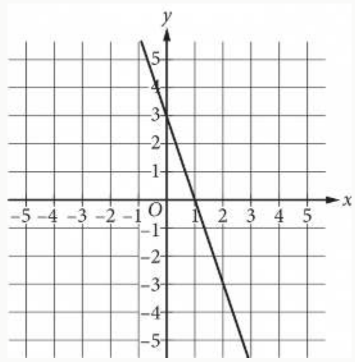
**B.** The number of minutes spent biking each day

**C.** The total number of minutes spent running and biking each day

**D.** The number of minutes spent biking for each minute spent running

Q9

ALGEBRA




What is the equation of the line shown in the xy-plane above?

A. $y = 3x - 3$

B. $y = -3x + 3$

C. $y = \frac{1}{3}x - 3$

D. $y = -\frac{1}{3}x + 3$

Q10 ALGEBRA

A city's total expense budget for one year was $x$ million dollars. The city budgeted $y$ million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between $x$ and $y$ in this context?

A. $x + y = 201$

B. $x - y = 201$

C. $2x - y = 201$

D. $y - x = 201$